

Shopping Mall Application

Project Problem Description

You have been requested to develop a shopping mall application with the following characteristics:

The CEO of a company plans to build a shopping mall application. Main users of the shopping mall application are CEO, staff, and customers. All users must be registered to use the shopping mall application. Customers can select items from the shopping mall and store them into wish list. However, for items that are sold out, customers cannot add them in the wish list. Customers can remove items from the wish list. Customers can pay by credit or debit card. Once purchase has been made, the system sends receipt to customer's registered email. Customers can send messages to staff in the shopping mall and Staff can reply to unread messages. Customers also can "like" the item. The number of "like" influences the order or appearance of the item in the shopping mall. Item that have high number of "like" will be show earlier than other items.

Staff is responsible of refilling inventories of items being sold in the shopping mall. Staff uploads or remove items sold in the shopping mall as well. Staff can write / modify item information. Staff has access to customer personal/payment information to provide answers from customers.

The system generates a daily summary report at 9 pm every business day and a monthly report at the end of each month. The report only can be accessed by CEO. It shows the information of company's sale for a day, week, and a month. The report contains each products name, number of items sold. The monthly report is a summary of daily reports for a month.

Course Project Score table

Course project is 30% of the total Grade. Student must submit total 3 deliverables for the project.

Course Project	30%
--Requirement Specification	5%
--Requirement Analysis	10%
--Architectural Design	5%
-- Implementation/Testing	10%

Individual Performance:

--For each submission, team members must submit individual team performance evaluation. Performance evaluation will affect the score you get for each submission. Please be true in writing performance of your teammates.

You are required to develop:

Requirements Specifications -Due Date will be announced in the class

- Develop the use case diagram for functionalities of shopping mall in terms of actors and use cases using the description above.
- Develop use case documentation for each use case you have developed using the template in the lecture note.

Requirements Analysis -Due Date will be announced in the class

- Use the use cases developed in requirement specification to develop the static model for the shopping mall, which depicts the classes and their relationships. A class should be classified as a boundary class, entity class, control class, or application logic class. Define the attributes of each entity class (roll of the class).
- Develop the communication model depicting objects participating in each use case and the sequence of interactions among the objects. A use case is modeled using a communication diagram.

Architectural Design -Due Date will be announced in the class

- Design the software architectural model for the shopping mall where the model is defined in terms of subsystems and their interactions. Each subsystem should be represented with objects supporting the subsystem.
- Define the communication styles between subsystems

Implementation/Testing -Due Date will be announced in the class

- Develop a running prototype. You need to implement the software using object-oriented language such as C++, C#, or Java. The prototype should have two functionalities designated by the instructor. GUI is **NOT** required.
- Design test cases that have test data and expected results. Testing must be conducted in all use cases to show how well your use cases work. Test should be conducted with correct inputs and incorrect inputs. You are required to turn in your test results with test cases.
- **Each team will set up a meeting with the instructor to demonstrate features of the prototype.**

Make your assumptions if necessary